

Com S 228 Recitation - Week 5

Sorting Algorithms, Stability and Time Complexity

Recitation 7

Iowa State University

February 15, 2019

- Exam 1 is **Wednesday 6:45-7:45**, February 20.
- Project 1 grading should be done by March 3.
- Project 2 is out, due March 3.

Table of Contents

- 1 Selection Sort
- 2 Bubble Sort
- 3 Insertion Sort
- 4 Quick Sort (Unstable)
- 5 Merge Sort
- 6 Counting Sort
- 7 Radix Sort

Selection Sort

Selection Sort

4, 3, 2, 7, 9, 1, 5, 6

1, 4, 3, 2, 7, 9, 5, 6

1, **2**, 4, 3, 7, 9, 5, 6

1, 2, **3**, 4, 7, 9, 5, 6

1, 2, 3, **4**, 7, 9, 5, 6

1, 2, 3, 4, **5**, 7, 9, 6

1, 2, 3, 4, 5, **6**, 7, 9

1, 2, 3, 4, 5, 6, **7**, 9

Bubble Sort

Bubble Sort

4, 3, 2, 7, 9, 1, 5, 6

1, 4, 3, 2, 7, 9, 5, 6

1, 2, 4, 3, 5, 7, 9, 6

1, 2, 3, 4, 5, 6, 7, 9

1, 2, 3, 4, 5, 6, 7, 9

1, 2, 3, 4, 5, 6, 7, 9

1, 2, 3, 4, 5, 6, 7, 9

1, 2, 3, 4, 5, 6, 7, 9

Selection Sort vs Bubble Sort

Selection Sort

4, 3, 2, 7, 9, 1, 5, 6

1, 4, 3, 2, 7, 9, 5, 6

1, **2**, 4, 3, 7, 9, 5, 6

1, 2, **3**, 4, 7, 9, 5, 6

1, 2, 3, **4**, 7, 9, 5, 6

1, 2, 3, 4, **5**, 7, 9, 6

1, 2, 3, 4, 5, **6**, 7, 9

1, 2, 3, 4, 5, 6, **7**, 9

Bubble Sort

4, 3, 2, 7, 9, 1, 5, 6

1, 4, 3, 2, 7, 9, 5, 6

1, **2**, 4, 3, 5, 7, 9, 6

1, 2, **3**, 4, 5, 6, 7, 9

1, 2, 3, **4**, 5, 6, 7, 9

1, 2, 3, 4, **5**, 6, 7, 9

1, 2, 3, 4, 5, **6**, 7, 9

1, 2, 3, 4, 5, 6, **7**, 9

Insertion Sort

Insertion Sort

4, 3, 2, 7, 9, 1, 5, 6

3, 4, ...

2, 3, 4, ...

2, 3, 4, 7, ...

2, 3, 4, 7, 9, ...

1, 2, 3, 4, 7, 9, ...

1, 2, 3, 4, 5, 7, 9, ...

1, 2, 3, 4, 5, 6, 7, 9

Quick Sort (Unstable)

Quick Sort

4, 3, 2, 7, 9, 1, 5, 6

4, 3, 2, 7, 9, 1, 5, 6

4, 3, 2, 7, 9, 1, 5, 6

4, 3, 2, 7, 9, 1, 5, 6

4, 3, 2, 7, 9, 1, 5, 6

4, 3, 2, 1, 9, 7, 5, 6

4, 3, 2, 1, 5, 7, 9, 6

4, 3, 2, 1, 5, 6, 7, 9

Merge Sort

Merge Sort

4, 3, 2, 7, 9, 1, 5, 6

3, 4, 2, 7, 1, 9, 5, 6

2, 4, 3, 7, 1, 9, 5, 6

2, 3, 4, 7, 1, 5, 9, 6

2, 3, 4, 7, 1, 5, 6, 9

2, 3, 4, 7, 1, 5, 6, 9

1, 3, 4, 7, 2, 5, 6, 9

1, 2, 4, 7, 3, 5, 6, 9

1, 2, 3, 7, 4, 5, 6, 9

1, 2, 3, 4, 7, 5, 6, 9

1, 2, 3, 4, 5, 7, 6, 9

1, 2, 3, 4, 5, 6, 7, 9

1, 2, 3, 4, 5, 6, 7, 9

Counting Sort

Assume keys are in the range of $[1, 5]$.

2, 1, 5, 3, 2, 1, 4, 2

Counting Sort

2, 1, 5, 4, 2, 1, 4, 2

- ① Take a count array to store the count of each unique object.

Index	1	2	3	4	5
Count	2	3	0	2	1

- ② Modify the count array such that each element at each index stores the sum of previous counts.

Index	1	2	3	4	5
Count	2	5	5	7	8

- ③ Output each object from the input sequence followed by decreasing its count by 1.

2, 1, 5, 4, 2, 1, 4, 2

2, 1, 5, 4, 2, 1, 4, 2

2, 1, 5, 4, 2, 1, 4, 2

Index	1	2	3	4	5
Count	2	5	5	7	8

Index	1	2	3	4	5
Count	2	4	5	7	8

Index	1	2	3	4	5
Count	1	4	5	7	8

				2			
--	--	--	--	---	--	--	--

	1			2			
--	---	--	--	---	--	--	--

	1			2			5
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Radix Sort

329, 457, 657, 839, 436, 720, 355

Radix Sort

3	2	9
4	5	7
6	5	7
8	3	9
4	3	6
7	2	0
3	5	5

7	2	0
3	5	5
4	3	6
4	5	7
6	5	7
3	2	9
8	3	9

7	2	0
3	2	9
4	3	6
8	3	9
3	5	5
4	5	7
6	5	7

3	2	9
3	5	5
4	3	6
4	5	7
6	5	7
7	2	0
8	3	9

Radix sort uses **counting sort** as a subroutine to sort.